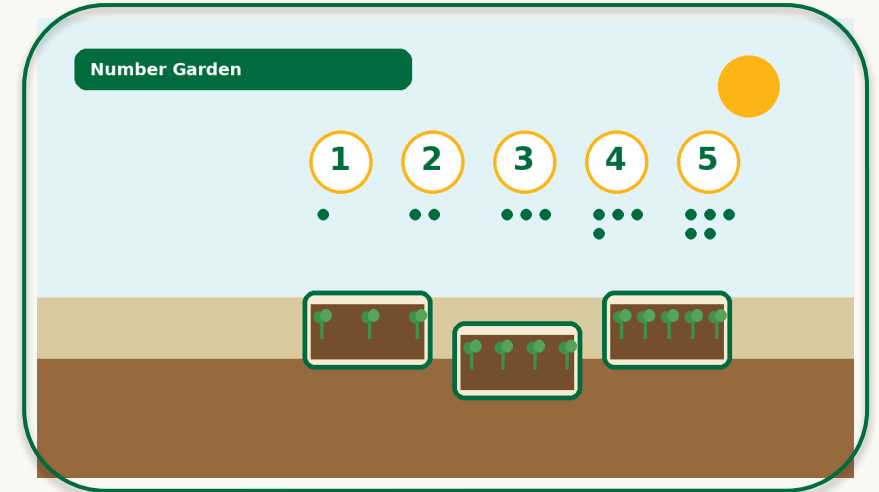


Number Garden

Early Numeracy Through Growing

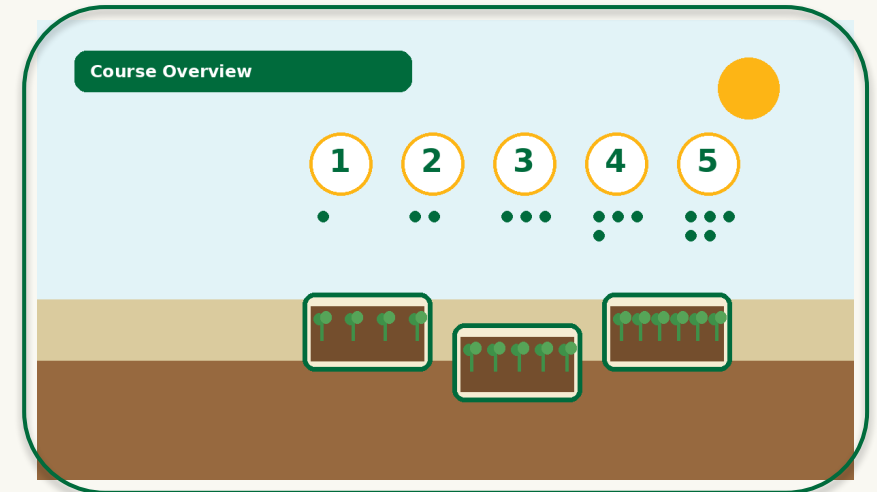
- LIFEWS AgriShine™ GrowMeal™ Teacher Training
- Audience: Nursery teachers and garden facilitators
- Format: Online micro-course + supervised garden practice
- Recommended duration: 2 hours 30 minutes online + 60 minutes practical



Course Overview

What teachers will learn and prepare

- Learning outcomes
- Explain how the Number Garden supports early numeracy.
- Teach counting using plants, seeds, leaves and garden objects.
- Connect written numerals with real quantities.
- Guide children to compare more, fewer and equal groups.
- Use patterns, grouping, planting and harvesting as numeracy experiences.
- Observe and document early numeracy learning.



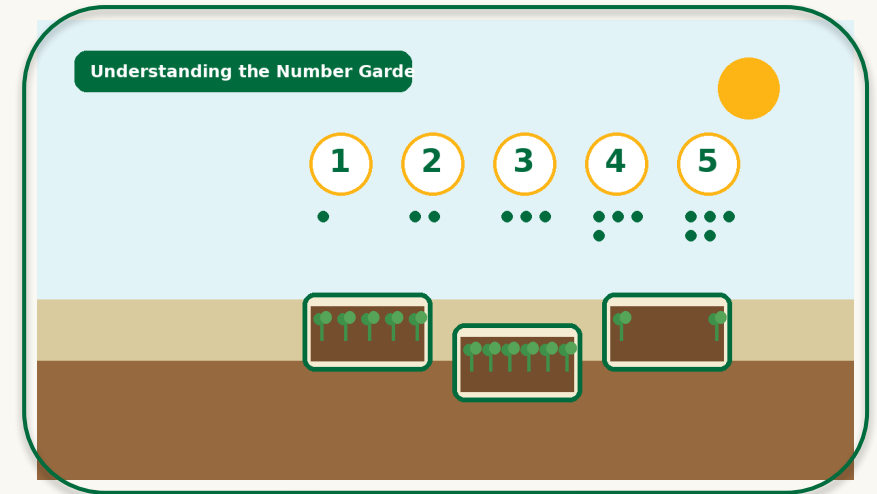
TEACHER PRACTICE

- Materials needed
- Number cards 0-10; counters; seeds; seedlings; leaves; small baskets; child-safe trowels; watering cans; sorting trays; picture cards; chalk/markers; simple observation checklist; handwashing supplies.

Understanding the Number Garden

A living mathematics environment

- A Number Garden turns real plants and garden materials into objects children can count, match, compare and organize.
- Begin with small quantities children can see clearly; increase complexity gradually.
- Use the same mathematical language consistently: number, count, how many, more, fewer, same/equal, group, pattern.
- Move from concrete objects to pictures and then to written numerals.
- Keep quantities unambiguous: separate objects so each plant, leaf or seed can be counted once.



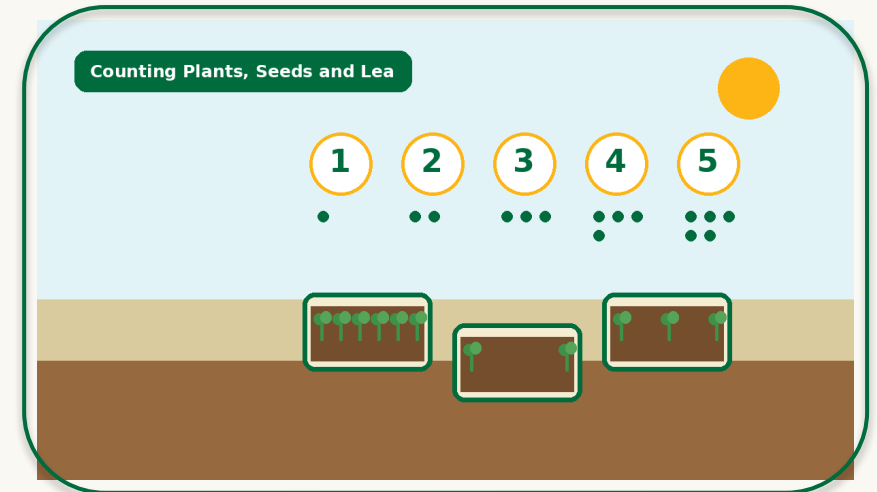
TEACHER PRACTICE

- Teacher practice: choose one bed and identify five different ways it could support counting.

Counting Plants, Seeds and Leaves

One object, one count

- Model one-to-one correspondence: touch or point to one object while saying one number word.
- Count slowly from left to right or use a clear physical path so objects are not counted twice.
- Ask children to count seeds before planting, leaves on selected plants, seedlings in a row and harvested items in a basket.
- Rearrange the same objects and count again to show that quantity remains the same.
- Encourage children to answer “How many?” after counting rather than restarting automatically.



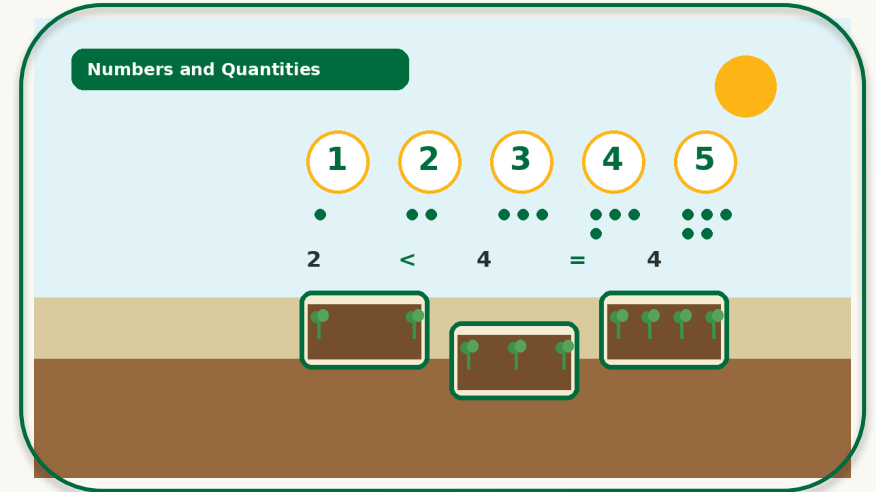
TEACHER PRACTICE

- Teaching sequence: POINT → COUNT → SAY HOW MANY → CHECK.

Numbers and Quantities

Connect the numeral to the amount

- Show a numeral card and ask children to build that quantity with safe garden objects.
- Reverse the task: show a group of objects, count them, then select the matching numeral.
- Use quantities 1-5 first; extend toward 10 as children become confident.
- Include zero naturally: an empty basket can represent 0 harvested items.
- Use different objects for the same number so children learn that a number describes quantity, not a specific object.



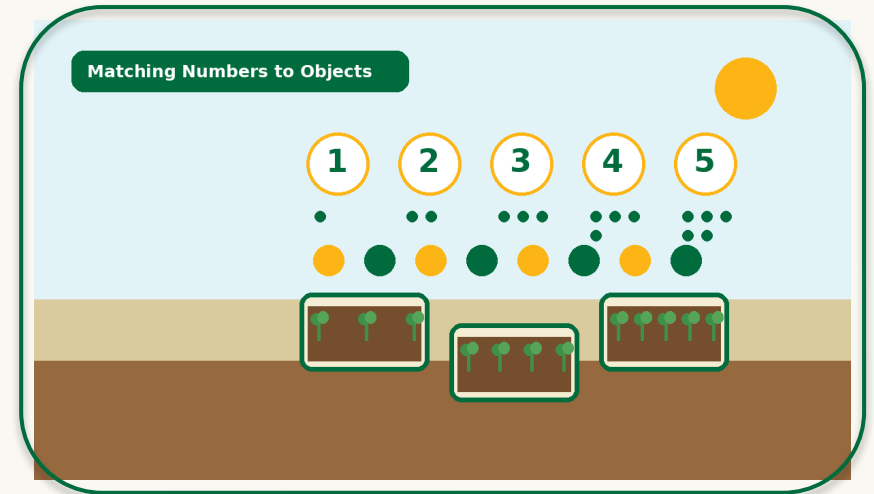
TEACHER PRACTICE

- Mini-activity: "Show me 4" using seeds, leaves, pots or seedlings.

Matching Numbers to Objects

Numerical ↔ quantity

- Place number cards beside groups of plants or objects and ask children to check whether the match is correct.
- Use baskets containing different numbers of safe items; children count and attach the matching card.
- Create garden stations for 1, 2, 3, 4 and 5 with clearly countable objects.
- Ask children to correct deliberate mismatches: a card showing 3 beside four leaves.
- Let children create their own numeral–quantity match for a partner to solve.



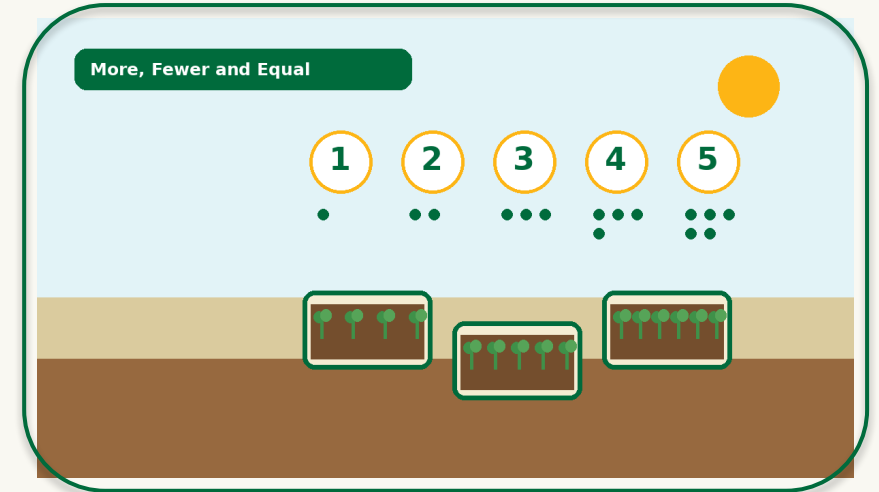
TEACHER PRACTICE

- Teacher check: the child should count the objects, not guess from the size of the group.

More, Fewer and Equal

Compare two quantities

- Begin with two clearly different groups and count both before comparing.
- Use precise language: "This group has more." "This group has fewer." "Both groups have the same number."
- Pair objects one-to-one to make comparisons visible.
- Progress from obvious differences, such as 2 and 5, to closer quantities, such as 4 and 5.
- Use real examples: seedlings in two rows, leaves in two trays or vegetables in two baskets.



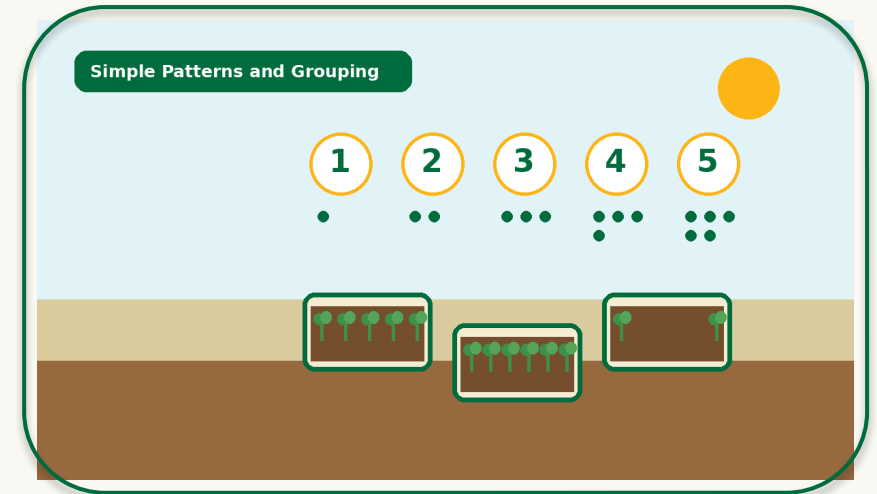
TEACHER PRACTICE

- Garden challenge: compare two groups, explain which has more/fewer, then make the groups equal.

Simple Patterns and Grouping

Notice order, repetition and sets

- Create repeating patterns with safe natural materials: leaf-seed-leaf-seed or green leaf-dry leaf-green leaf-dry leaf.
- Ask children to identify what comes next and explain the repeating unit.
- Group objects by type, size, color or another visible feature, then count each group.
- Introduce equal groups with simple arrangements such as two groups of three seeds.
- Use planting layouts to show rows and repeated spacing without introducing formal multiplication.



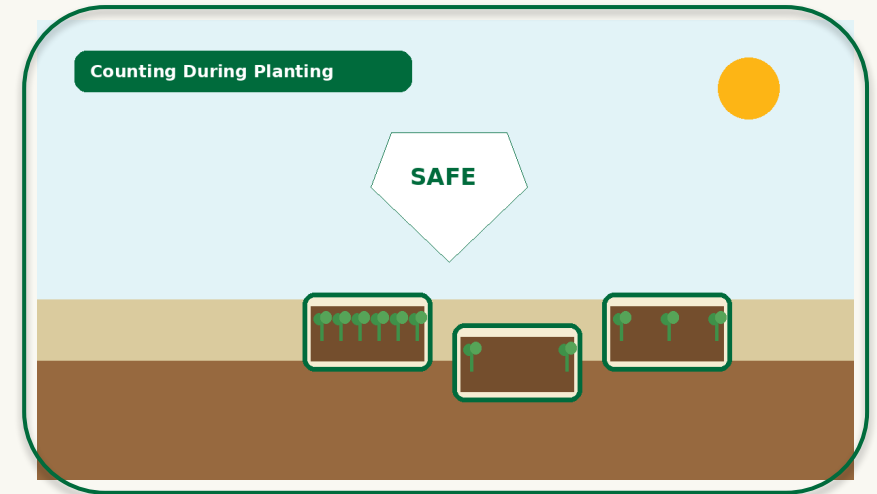
TEACHER PRACTICE

- Teacher prompt: "What is the pattern? What comes next? How many are in each group?"

Counting During Planting

Numeracy becomes a real garden task

- Count the number of planting holes, seeds, seedlings and watering turns.
- Give children a target quantity: plant 3 seeds, place 4 seedlings or water 5 pots.
- Use number cards at planting stations to show the required quantity.
- Count together before planting and check again after planting.
- Keep seeds large enough to handle safely and follow school rules on materials appropriate for young children.
- Connect action words with number language: first, next, last, one more, one fewer.



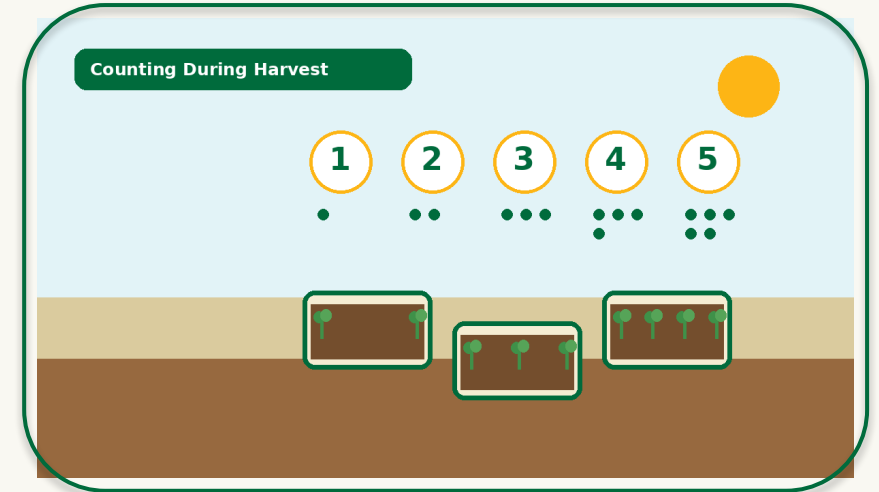
TEACHER PRACTICE

- Practical routine: COUNT → PLANT → RECOUNT → LABEL.

Counting During Harvest

From garden quantity to simple records

- Harvest only produce identified as ready and approved by the supervising teacher.
- Count harvested items into small baskets or trays.
- Compare two baskets: which has more, fewer or the same?
- Sort produce into simple groups before counting.
- Match each basket with a numeral card representing its quantity.
- Record the result with a drawing, tally-like marks appropriate to the class, or a teacher-written numeral.



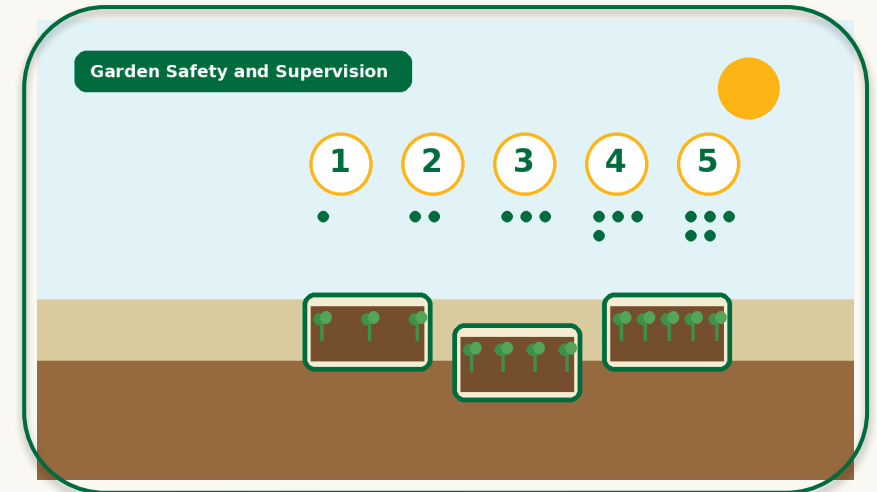
TEACHER PRACTICE

- Keep the learning focus on quantity; avoid making speed or competition the main goal.

Garden Safety and Supervision

Safe numeracy is planned numeracy

- Inspect paths, beds, tools, water points and plants before pupils enter.
- Use child-safe tools and materials; supervise seeds and small objects according to school age-safety rules.
- Establish clear boundaries and a gathering point.
- Use small groups and stations so every child can count without crowding.
- Teach: walk on paths, use tools when instructed, ask before touching/tasting, wash hands after gardening.
- Know school procedures for allergies, bites/stings, heat, weather and first aid.



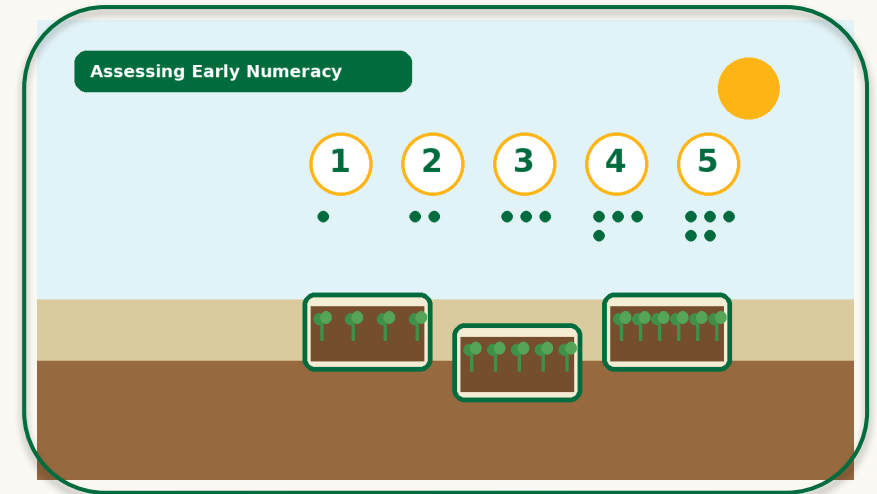
TEACHER PRACTICE

- Pre-lesson check: PEOPLE • PLACE • MATERIALS • WATER • WEATHER • HYGIENE.

Assessing Early Numeracy

Observe what children can do in the garden

- Use brief observation during authentic activities rather than relying only on worksheets.
- Look for: recites number sequence; counts objects one-to-one; answers “how many”; matches numeral to quantity; compares more/fewer/equal; continues a simple pattern; forms a requested group.
- Record whether the child performs independently, with a prompt or not yet.
- Collect simple evidence: teacher checklist, child-created numeral-quantity match, drawing or short oral response.
- Reassess across several garden visits because early numeracy develops through repeated experiences.



TEACHER PRACTICE

- Completion task: design one 30-minute Number Garden lesson using counting, comparison and one practical garden activity.
- Teacher readiness: explain the garden, organize safe stations, model accurate counting language, correct common counting errors and document learning.